Sarah Woodcock – Machine Learning Module Critical Reflection

E-Portfolio: https://sosswood.github.io/Machine-Learning/

Description

I have been on an extremely steep learning curve with this module. I have learned about Github and how to use it. I have learned about Python and Google Colab as well as learning the techniques relating to Machine Learning. All of these things were entirely new to me.

In Unit 6 I worked on a group project with three other team members. The project was related to regression and clustering, using a New York Airbnb dataset. My main responsibilities in the group project were administrative. I organised most of the meetings, put everyone's contributions together in a final, cohesive document and, after everyone had approved it, I submitted the final report.

In Unit 11, I developed recommender systems – after a few false starts with different datasets. I created these using traditional matrix factorisation and with neural collaborative filtering and then evaluated both.

Throughout the module there has been emphasis on not just the practical work, but also considering the efficacy of different techniques as well as the ethical implications of their application.

Feelings

The group project in Unit 6 was daunting. My submission was dependent on three strangers. Those strangers are now colleagues and we worked really well together and supported each other throughout.

At the start, I felt I was far less confident that the other team members in my abilities to carry out the technical work, which is why I offered to take on the admin side. I doubted that I could contribute as effectively as everyone else.

However, the supportiveness of my team and the allocation of tasks that meant most of my contribution came towards the end of the project, gave me time to experiment with clustering and regression on my

own, without the pressure that my work needed to be correct and completed within the project timescales. This meant I could work to my strengths whilst improving my weaknesses.

For my Individual Project, I still struggled with knowing which bits of Python code to use in each circumstance. I felt I spent more time trying to find things in the textbook than actually running the code. I often had to ask Gemini why my code hadn't worked, but these were normally small corrections. I understood what I needed to do, but I experienced the frustration of knowing that, if I was just more familiar with Python, I could do things much more quickly.

Evaluation

This is a master's level course and I knew it would be difficult. There has been an enormous amount to learn in a short space of time. Along with a full-time job, moving my daughter across the country to start university and some illness, there were times I nearly gave up. There have been tears and tantrums.

The support of my team members from Unit 6 was amazing. I used the online Chat feature once to check that I had correctly submitted my Individual Project as I really wasn't sure and the lady that responded to me was so helpful.

Our lecturer was engaging and fun. Unfortunately, I wasn't available to attend any of the seminars in person, but the online course allows for this and I was able to watch them all at a later date.

Analysis

I started out thinking that I would learn about Machine Learning techniques and perhaps learn a small amount of Python. I've learned so much more!

I wanted to have a deeper understanding of Machine Learning so that I could use this in my work, but I've developed skills in Python and an understanding of GitHub that I'm already using in my work. I often have to analyse customer data for missing information and used to do this in Excel, by using filters. I now pop it into Colab and use some EDA techniques to see what's missing and what's duplicated.

I have progressed from using regression and clustering to developing a recommender system using both matrix factorisation (MF) and neural collaborative filtering (NCF) techniques. I am so impressed by the NCF models that I want to explore these further. I was disappointed that this model wasn't able to give better results than the traditional model in the time I had available.

I was incredibly lucky with my group allocation. I learned a lot from my team as they were able to talk knowledgeably about our findings to date during our team meetings and they came up with great ideas.

The Individual Project was really challenging for me. I was constantly on the edge of being out of my depth. In signing up for this module, having last studied at university level in 1998, when everything I

submitted was either hand-written, or printed out as double-line spaced and physically handed to the department office, completing a Machine Learning module was an unbelievably different experience. I'm grateful that I've had the opportunity to do this.

Conclusion

Looking back at the last twelve weeks, I can see that I've learned many technical skills, which is what I expected. However, I've also learned about myself. I shouldn't be so full of doubt or underestimate myself. I have strengths such as perseverance, team leadership and collaboration which I brought to this module and should apply more in my daily life. It's good to push myself out of my comfort zone.

Action Plan

I'm going to carry on trying to improve my Python skills. I will probably run through the course's Jupyter Notebooks again – a little bit more leisurely this time – and also re-read the text book. I want basic Python code to be as second-nature to me as Excel formulae.

I will continue to use Python in my daily work and hopefully use my new-found knowledge regarding evaluating system behaviour and understanding the implications of automation to assist my company in its quest to implement Machine Learning and AI into our products.

I'd love to complete the MSc Artificial Intelligence programme at some point in the future, funds and time permitting.

References

Gibbs, G. (1988) *Learning by doing: a guide to teaching and learning methods*. Oxford: Further Education Unit, Oxford Polytechnic.

University of Edinburgh (n.d.) Reflecting on experience. Available at: https://reflection.ed.ac.uk/reflectors-toolkit/reflecting-on-experience (Accessed: 17 October 2026).